MATH317 · NUMERICAL METHODS

Midterm 472 Solutions

12/04/2026 · 60 minutes · 30 marks

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01 Solution Q1: True / False

#	Answer	Reason
1	F	Digits matter throughout the calculation. Rounding in the middle grows the error.
2	T	That is the purpose of Horner: evaluating a polynomial without powers.
3	F	It converges rapidly near the expansion point, not far from it.
4	F	It gives an approximation that improves each step, not the exact root.
5	T	This is the difference between rounding and chopping.
6	F	It is the case `n = 0` of Taylor's theorem, not "zero derivatives".
7	F	The correct bound is $\frac{b-a}{2^{n+1}}$.

02 Solution Q2: Horner and Taylor

Order the function from the highest power: `3x³ − 4x² + 0x + 22`. Coefficients:

`3, −4, 0, 22`.

```
a)        3     −4      0     22
 −2 |           −6     20    −40
          3    −10     20    −18
```

P(−2) = −18

```
b)        3     −4      0     22
 −3 |           −9     39   −117
          3    −13     39 |  −95        c0 = −95
               −9     66
          3    −22 |  105              c1 = 105
               −9
          3 |  −31                     c2 = −31
          3                           c3 = 3
```

P(x) = 3(x+3)³ − 31(x+3)² + 105(x+3) − 95

Because `c = −3`, the bracket is `(x − (−3)) = (x + 3)`. The key also accepts the solution by successive derivatives.

Marks in the key	Marks
(a) the table	2
(a) writing `P(−2) = −18`	1
(b) each of the three rows	1 + 1 + 1
(b) writing the final expansion	2

03 Solution Q3: Floating-point

1. $(23)_{10} = (27)_8 = (10111)_2$ [1 mark]

2. $(.21875)_{10} = (.16)_8 = (.001110)_2$ [1 mark]

3. $(23.21875)_{10} = (10111.001110)_2 = (1.0111001110)_2 \times 2^4$ [1 mark]

   stored mantissa: .0111001110

4. $c - 127 = 4 \quad \rightarrow \quad c = 131 = (203)_8 = (1000\ 0011)_2$ [2 marks]

5. $[1\ 10000011\ 0111001110...0000]_2$ [1 mark]

   $[1100\ 0001\ 1011\ 1001\ 1100\ ...\ 0000]_2$

   $= [\mathbf{C1B9C000}]_{16}$ [1 mark]

Every line carries a mark, so do not skip any.

04 Solution Q4: Bisection

a) $u = f(0) = -1 \qquad v = f(1) = -1 + 4 - e = 0.2817$

   $u \cdot v < 0 \quad \rightarrow \quad f$ has a root α in $[0, 1]$ [2 marks]

b) $n = 0$: $x_0 = \frac{0 + 1}{2} = 0.5$ $\quad w = f(0.5) = 0.1013$

$u \cdot w < 0 \quad \rightarrow \quad a = a\ ,\ b = x_0 \quad \rightarrow \quad [0, 0.5]$

$n = 1$: $x_1 = \frac{0 + 0.5}{2} = 0.25$ $\quad w = f(0.25) = -0.3465$

$u \cdot w > 0 \quad \rightarrow \quad a = x_1\ ,\ b = b \quad \rightarrow \quad [0.25, 0.5]$

$n = 2$: $x_2 = \frac{0.25 + 0.5}{2} = 0.375$ $\quad w = f(0.375) = -0.0956$

$u \cdot w > 0 \quad \rightarrow \quad a = x_2\ ,\ b = b \quad \rightarrow \quad [0.375, 0.5]$

n	x_n	$f(x_n)$	Error
0	0.5	0.1013	0.5
1	0.25	−0.3465	0.25
2	0.375	−0.0956	0.125

Each iteration is worth 2 marks.

Notes on the key

The key prints `x₂ = 2.75` in the table and an error of `0.25` for the third iteration in the text. Both are typos; the correct values are `0.375` and `0.125`.

The key uses only the sign change. To show the root is unique: `f″(x) = −2 − eˣ < 0`, so `f′` decreases, with `f′(0) = 3 > 0` and `f′(1) ≈ −0.72 < 0`. The function rises then falls, and while falling it stays above `f(1) = 0.28 > 0`. So it crosses the axis exactly once.